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Assignment 1: Mlops Apache Pig – Product Review Analysis

Title

Apache Pig for Product Review Data Processing and Average Rating Analysis

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To use Apache Pig and Pig Latin to process an e-commerce product review dataset stored in Hadoop HDFS.

Objective

- To load the **e-commerce product review dataset** from HDFS using Apache Pig.
- To identify and **filter out invalid reviews** with ratings below **1.0**.
- To remove records containing **null or missing rating values**.
- To **group valid reviews by product category**.
- To calculate the **average rating score** for each product category.
- To use **Pig Latin procedural dataflow operations** such as LOAD, FILTER, GROUP, and AVG.
- To generate meaningful category-wise rating results for analyzing **customer reviews and product performance**.

Problem Statement

An e-commerce company has a dataset containing product reviews and ratings.

Some reviews contain invalid ratings. A rating is considered invalid when:

- The rating is null.

- The rating is below 1.0.

The task is to use Apache Pig to remove these invalid reviews and calculate the average rating for each product category.

The final result should provide the average rating of every category containing valid reviews

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Technologies Used

Apache Hadoop	Distributed data processing framework
HDFS	Storage of the review dataset and output
YARN	Resource and job management
Apache Pig 0.17.0	| Data processing using Pig Latin
Pig Latin	Data-flow scripting language
Linux/Ubuntu	Operating environment

Hadoop Environment

Before executing the Pig program, the Hadoop services were started.

The following command was used to verify the running services:

bash
>> Jps

The following Hadoop processes were running:

NameNode
DataNode
SecondaryNameNode
ResourceManager
NodeManager

This confirms that both HDFS and YARN were running successfully.
Apache Pig Verification

Apache Pig version 0.17.0 was installed and verified using:

>> Pig -version

The installed version was:

Apache Pig version 0.17.0

Pig was also started successfully using:

Pig

Pig selected MapReduce as the execution engine and successfully connected to:

Hdfs://localhost:9000

This confirms that Apache Pig was correctly integrated with Hadoop.

Input Dataset

The input file used for the experiment is:

Reviews.csv

The dataset contains the following fields:

Field	***Description***
Review_id	Unique ID of the review
Product_id	Unique ID of the product
Category	Product category
Rating	Rating given by the customer

The dataset used for the experiment is:

Review_id,product_id,category,rating
1,P101,Electronics,4.5
2,P102,Books,3.0
3,P103,Electronics,5.0
4,P104,Books,0.0
5,P105,Electronics,
6,P106,Books,4.0
7,P107,Clothing,2.5
8,P108,Clothing,0.0
9,P109,Electronics,3.5

Invalid Records

The dataset contains invalid records such as:

4,P104,Books,0.0
5,P105,Electronics,

8,P108,Clothing,0.0

These records are removed because:

0.0 is below the minimum valid rating of 1.0.

An empty rating is a null value.

Storing the Dataset in HDFS

An HDFS directory was created using:

>> hdfs dfs -mkdir -p /user/aiml/reviews

The input file was uploaded to HDFS using:

>> Hdfs dfs -put -f reviews.csv /user/aiml/reviews/

The contents of the HDFS directory were verified using:

>> Hdfs dfs -ls /user/aiml/reviews

The input file was then displayed using:

>> Hdfs dfs -cat /user/aiml/reviews/reviews.csv

The HDFS input location is:

/user/aiml/reviews/reviews.csv

Pig Latin Program

The Pig Latin program is stored in:

Review.pig

The complete program is:

```
reviews = LOAD '/user/aiml/reviews/reviews.csv'
USING PigStorage(',')
AS (review_id:int, category:chararray, rating:double);
```

```
valid_reviews = FILTER reviews
BY rating IS NOT NULL AND rating >= 1.0;

grouped_reviews = GROUP valid_reviews BY category;

average_ratings = FOREACH grouped_reviews
GENERATE group AS category,
AVG(valid_reviews.rating) AS average_rating;

STORE average_ratings INTO '/user/aiml/review_average_output'
USING PigStorage(',');
```

Explanation of the Pig Program

1 Loading the Dataset

The following statement loads the dataset from HDFS:

```
Reviews = LOAD '/user/aiml/reviews/reviews.csv'
USING PigStorage(',')
AS (
Review_id:int,
Product_id:chararray,
Category:chararray,
Rating:float );
```

LOAD reads the data from HDFS.
PigStorage(',') specifies that the fields are separated by commas.
The schema defines the data types of the four fields.

2 Removing the Header

The following statement is used to remove the header row:

```
Data = FILTER reviews BY review_id IS NOT NULL;
```

The header does not contain an integer review_id, so it is removed by this filter.

3 Removing Invalid Ratings

The main filtering operation is:

```
Valid_reviews = FILTER data
BY rating IS NOT NULL AND rating >= 1.0;
```

This performs two checks:

Rating IS NOT NULL removes reviews without a rating.
Rating >= 1.0 removes ratings below 1.0.
Therefore, only valid ratings are retained.

4 Grouping by Category

The valid records are grouped using:
Grouped_reviews = GROUP valid_reviews BY category;
This creates one group for each product category.
For this dataset, the categories are:
Books
Clothing
Electronics

5 Calculating Average Rating

The average rating is calculated using:
Average_ratings = FOREACH grouped_reviews GENERATE Group AS category,
AVG(valid_reviews.rating) AS average_rating;
AVG() calculates the average rating for each category.

6 Storing the Result

The final results are stored in HDFS using:
STORE average_ratings INTO '/user/aiml/review_average_output'
USING PigStorage(',');
The output directory is:
/user/aiml/ review_average_output'

Executing the Pig Program

The Pig program was executed using:
>> Pig review_analysis.pig
Pig selected MapReduce as the execution engine.
The program successfully completed with:
MapReduceLauncher – Success!
This confirms that the Pig data-processing job was successfully executed.

Output Verification

The generated output directory was checked using:
>> Hdfs dfs -ls /user/aiml/review_output
The output contained:
_SUCCESS
Part-r-00000

The final output was displayed using:
>> Hdfs dfs -cat /user/aiml/review_output/part-r-00000
The output was:
Books,3.5
Clothing,2.5
Electronics,4.333333333333333

Result Analysis Books

Valid ratings:
3.0
4.0
Average:
(3.0 + 4.0) / 2 = 3.5
Therefore:
Books = 3.5
Clothing
Valid rating:
2.5
The rating 0.0 was removed because it is below 1.0.
Average:
2.5
Therefore:
Clothing = 2.5
Electronics
Valid ratings:
4.5
5.0
3.5
The null rating was removed.
Average:

(4.5 + 5.0 + 3.5) / 3
= 13.0 / 3
= 4.333333333333333
Therefore: Electronics = 4.333333333333333

Commands used:



```
2026-09-25 14:04:59,410 [main] INFO  org.apache.pig.Main - Pig script completed in 2 minutes, 49 seconds and 1
aiml@aiml-d204-13:~$ hdfs dfs -ls /user/aiml/review_average_output
2026-09-25 14:05:40,358 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform...
Found 2 items
-rw-r--r--   1 aiml supergroup          0 2026-09-25 14:02 /user/aiml/review_average_output/_SUCCESS
-rw-r--r--   1 aiml supergroup         50 2026-09-25 14:02 /user/aiml/review_average_output/part-r-00000
aiml@aiml-d204-13:~$ hdfs dfs -cat /user/aiml/review_average_output/part-r-00000
2026-09-25 14:05:53,136 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform...
Home,2.5
Books,3.75
Clothing,4.5
Electronics,3.75
aiml@aiml-d204-13:~$ hdfs dfs -get /user/aiml/review_average_output/part-r-00000 ~/pig-assignment/output.txt
2026-09-25 14:06:01,939 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform...
get: `/home/aiml/pig-assignment/output.txt': File exists
aiml@aiml-d204-13:~$ cat ~/pig-assignment/output.txt
Home,2.5
Books,3.75
Clothing,4.5
Electronics,3.75
aiml@aiml-d204-13:~$
```

(1) WhatsApp — Moz... | pig-assignment | aiml@aiml-d204-13: ~ | aiml@aiml-d204-13: ~ | output.txt (~/pig-assi...

Output in textfile

output.txt

```
1 Home,2.5
2 Books,3.75
3 Clothing,4.5
4 Electronics,3.75
```

Reviews.csv

	A	B	C
1	Electronics	4.5
2	Electronics	3
3	Electronics	0.5
4	Clothing	5
5	Clothing	4
6	Clothing	0
7	Books	4
8	Books	3.5
9	Books	
10	Home	2.5